| EXP.NO: | 5 | CPU SCHEDULING ALGORITHMS (FCFS, SJF, PRIORITY, ROUND ROBIN) |
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## AIM

To implement and compare CPU scheduling algorithms: First Come First Served (FCFS), Shortest Job First (SJF), Priority Scheduling, and Round Robin (RR).

## PROGRAM 1 – First Come First Served (FCFS)

```
#include <stdio.h>
int main() {
    int n, i;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int AT[n], BT[n], CT[n], TAT[n], WT[n];
    for(i=0;i<n;i++) {
        printf("Enter AT and BT for P%d: ", i+1);
        scanf("%d %d", &AT[i], &BT[i]);
    }
    CT[0] = AT[0] + BT[0];
    for(i=1;i<n;i++)
        CT[i] = (CT[i-1] < AT[i]) ? AT[i]+BT[i] : CT[i-1]+BT[i];
    float totalTAT=0, totalWT=0;
    for(i=0;i<n;i++) {
        TAT[i]=CT[i]-AT[i]; WT[i]=TAT[i]-BT[i];
        totalTAT+=TAT[i]; totalWT+=WT[i];
    }
    printf("\nAvg TAT: %.2f\n", totalTAT/n);
    printf("Avg WT:  %.2f\n", totalWT/n);
    return 0;
}
```

## PROGRAM 2 – Shortest Job First (SJF)

```
#include <stdio.h>
int main() {
    int n,i,time=0,completed=0,min;
    printf("Enter number of processes: "); scanf("%d",&n);
    int AT[n],BT[n],CT[n],TAT[n],WT[n],visited[n];
    for(i=0;i<n;i++){printf("Enter AT,BT for P%d: ",i+1);scanf("%d
%d",&AT[i],&BT[i]);visited[i]=0;}
    while(completed<n){
        min=-1;
        for(i=0;i<n;i++) if(AT[i]<=time&&visited[i]==0) if(min==-1||BT[i]<BT[min]) min=i;
        if(min==-1) time++; else {time+=BT[min];CT[min]=time;visited[min]=1;completed++;}
    }
    float tTAT=0,tWT=0;
    for(i=0;i<n;i++){TAT[i]=CT[i]-AT[i];WT[i]=TAT[i]-BT[i];tTAT+=TAT[i];tWT+=WT[i];}
    printf("\nAvg TAT: %.2f\nAvg WT: %.2f\n",tTAT/n,tWT/n);
    return 0;
}
```

## PROGRAM 3 – Priority Scheduling

```
#include <stdio.h>
int main() {
    int n,i,time=0,completed=0,min;
    printf("Enter number of processes: "); scanf("%d",&n);
    int AT[n],BT[n],P[n],CT[n],TAT[n],WT[n],visited[n];
    for(i=0;i<n;i++){
        printf("Enter AT,BT,Priority for P%d: ",i+1);
        scanf("%d %d %d",&AT[i],&BT[i],&P[i]); visited[i]=0;
    }
    while(completed<n){
        min=-1;
        for(i=0;i<n;i++) if(AT[i]<=time&&visited[i]==0) if(min==-1||P[i]<P[min]) min=i;
        if(min==-1) time++; else {time+=BT[min];CT[min]=time;visited[min]=1;completed++;}
    }
    float tTAT=0,tWT=0;
    for(i=0;i<n;i++){TAT[i]=CT[i]-AT[i];WT[i]=TAT[i]-BT[i];tTAT+=TAT[i];tWT+=WT[i];}
    printf("\nAvg TAT: %.2f\nAvg WT: %.2f\n",tTAT/n,tWT/n);
    return 0;
}
```

## PROGRAM 4 – Round Robin

```
#include <stdio.h>
int main() {
    int n,tq,i,time=0,completed=0;
    printf("Enter number of processes: "); scanf("%d",&n);
    printf("Enter Time Quantum: "); scanf("%d",&tq);
    int AT[n],BT[n],RT[n],CT[n],TAT[n],WT[n];
    for(i=0;i<n;i++){printf("Enter AT,BT for P%d: ",i+1);scanf("%d
%d",&AT[i],&BT[i]);RT[i]=BT[i];}
    while(completed<n){
        int done=1;
        for(i=0;i<n;i++){
            if(RT[i]>0&&AT[i]<=time){done=0;
                if(RT[i]>tq){time+=tq;RT[i]-=tq;}
                else{time+=RT[i];CT[i]=time;RT[i]=0;completed++;}
            }
        }
        if(done) time++;
    }
    float tTAT=0,tWT=0;
    for(i=0;i<n;i++){TAT[i]=CT[i]-AT[i];WT[i]=TAT[i]-BT[i];tTAT+=TAT[i];tWT+=WT[i];}
    printf("\nAvg TAT: %.2f\nAvg WT: %.2f\n",tTAT/n,tWT/n);
    return 0;
}
```

## OUTPUT

```
--- FCFS ---
$ gcc fcfs.c && ./a.out
Enter number of processes: 3
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 1 3
Enter AT and BT for P3: 2 8
Avg TAT: 9.00
Avg WT:  5.67

--- SJF ---
$ gcc sjf.c && ./a.out
Enter number of processes: 3
Enter AT,BT for P1: 0 8
Enter AT,BT for P2: 1 4
Enter AT,BT for P3: 2 2
Avg TAT: 7.33
Avg WT:  4.00

--- PRIORITY ---
$ gcc priority.c && ./a.out
Enter number of processes: 3
Enter AT,BT,Priority for P1: 0 5 3
Enter AT,BT,Priority for P2: 0 3 1
Enter AT,BT,Priority for P3: 0 8 2
Avg TAT: 9.33
Avg WT:  5.67

--- ROUND ROBIN ---
$ gcc rr.c && ./a.out
Enter number of processes: 3
Enter Time Quantum: 2
Enter AT,BT for P1: 0 5
Enter AT,BT for P2: 0 4
Enter AT,BT for P3: 0 3
Avg TAT: 10.00
Avg WT:  6.33
```

## RESULT

CPU scheduling algorithms — FCFS, SJF, Priority, and Round Robin — were implemented in C and tested with sample inputs. Average Turnaround Time and Waiting Time were computed successfully.